ClaimGuard

Span-Level Hallucination Detection for AI Text

Client overview · Prepared for sharing · Raunak Rathi

1. Executive Summary

ClaimGuard checks AI-generated summaries or answers against a source document. It splits the generated text into sentences and labels each span as Supported, Contradicted, or Baseless. Users get a clear hallucination rate, overall verdict, and color-highlighted spans.

Primary users: Compliance, finance reporting, RAG product teams validating groundedness.

2. Key Features

- Paste source document and generated text
- Sentence-level span classification
- Labels: Supported / Contradicted / Baseless
- Overall support score and hallucination rate
- Color-coded highlight view in the dashboard
- Verification history and upcoming-feature roadmap

3. How the System Works (Code Flow)

The application follows a simple layered design: React dashboard → FastAPI backend → core processing module → SQLite storage.

1. User pastes source text and generated text (or uses sample files).
2. Frontend posts both texts to the verification API.
3. Backend splits the generated text into sentence spans.
4. For each span, the best matching evidence in the source is found.
5. NLI (or demo heuristics) classifies the span against that evidence.
6. Scores are aggregated into hallucination rate and a verdict.
7. Results are stored and returned for UI highlighting.

4. Technology Stack

- Frontend: React + Vite + Tailwind (port 5172)
- Backend: FastAPI + SQLite (port 8002)
- AI: DeBERTa-style NLI + embeddings; demo mode without GPU

5. How to Run

```
cd claimguard/backend && DEMO_MODE=1 uvicorn main:app --reload --port 8002
cd claimguard/frontend && npm install && npm run dev
Open http://127.0.0.1:5172
```

6. Sample Files for Demo

- samples/source_report.txt — grounded earnings excerpt
- samples/generated_summary.txt — summary with planted errors
- samples/verification_pair.json — same pair for API testing
- Planted issues include polarity flips and unsupported claims

7. Main API Endpoints

- POST /api/verify — run span verification
- GET /api/history — past verifications
- GET /api/stats — dashboard metrics
- GET /api/verification/{id} — fetch one result

8. Upcoming Features

The following capabilities are planned for future releases and are shown in the product UI under Upcoming.

- Black-box detector baselines (consistency / LLM-as-judge)
- Fine-tuning on public hallucination corpora
- Abstain or regenerate when risk is high
- Deeper failure-mode analysis by error type

9. Folder Layout

- claimguard/frontend — web dashboard
- claimguard/backend — API and core logic
- claimguard/samples — demo upload / paste files
- claimguard/docs — internal report and slides